AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956
SCHOOL OF COMPUTING CHENNAI

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

23CSE304 Embedded Systems Laboratory

Activity Sheet #9

Activity Name: Timer Programming

Name of Student:	**Agneay B Nair**
Registration No.:	**CH.SC.U4CSE24102**
Activity Date:	**21-08-26**
Submission Date:	**21-08-26**

Assessment Rubrics:

Component	Maximum Marks	Marks Allotted
Program	3	
Output	4	
Viva	2	
Timely Submission	1	
Total	**10**	

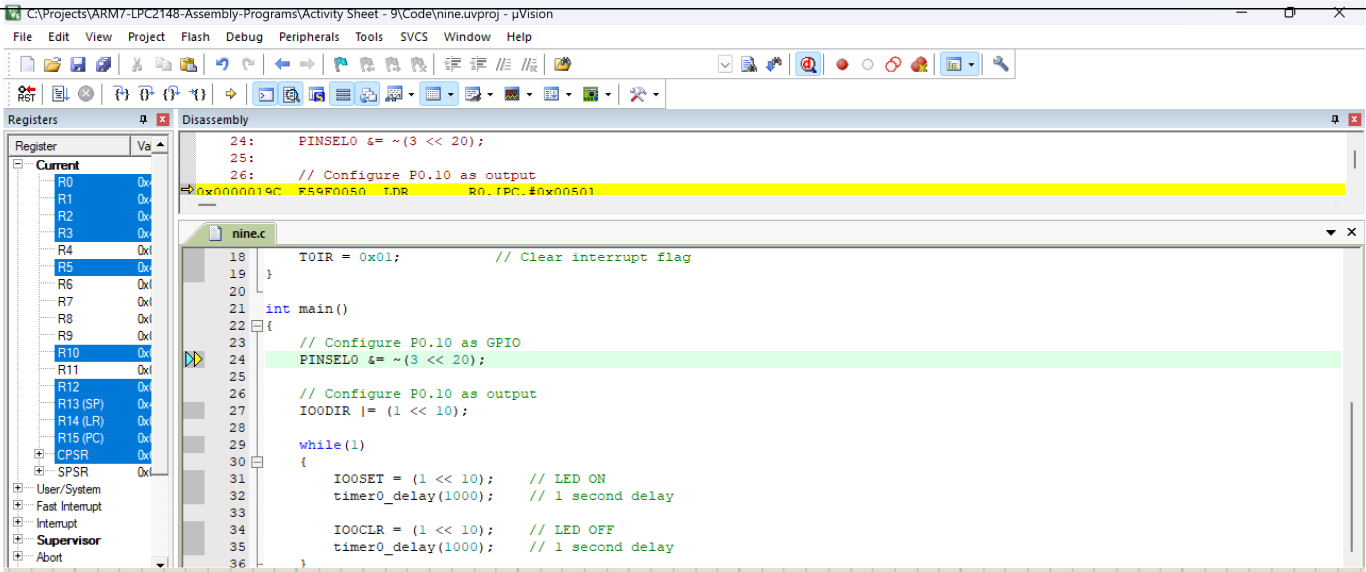
Faculty In-charge

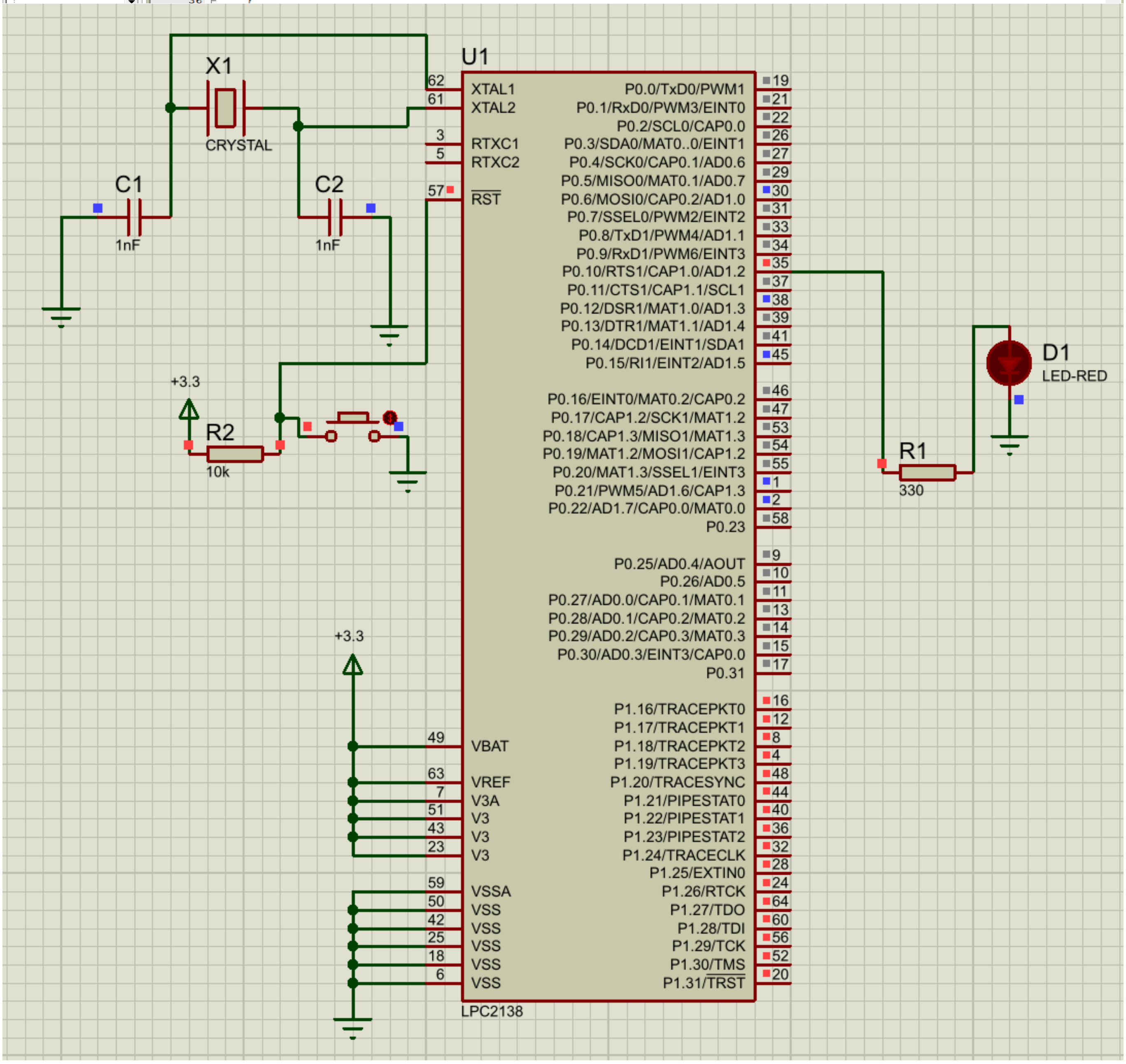
Write an embedded C program for ARM architecture to drive Timer module. Debug and verify the program using IDE on a development computer system.

CODE:

```
#include <LPC214x.h>

void timer0_delay(unsigned int delay_ms)
{
   T0TCR = 0x02;         // Reset Timer0

   T0PR = 14999;         // Prescaler for 1 ms delay
   T0TC = 0;             // Clear Timer Counter

   T0MR0 = delay_ms;     // Match value

   T0MCR = 0x04;         // Stop timer when TC = MR0

   T0TCR = 0x01;         // Start Timer0

   while(!(T0IR & 0x01)); // Wait until match occurs

   T0IR = 0x01;          // Clear interrupt flag
}

int main()
{
   // Configure P0.10 as GPIO
   PINSEL0 &= ~(3 << 20);

   // Configure P0.10 as output
   IO0DIR |= (1 << 10);

   while(1)
   {
      IO0SET = (1 << 10);    // LED ON
      timer0_delay(1000);    // 1 second delay

      IO0CLR = (1 << 10);    // LED OFF
      timer0_delay(1000);    // 1 second delay
   }
}
```

OUTPUT:


C:\Projects\ARM7-LPC2148-Assembly-Programs\Activity Sheet - 9\Code\nine.uvproj - µVision
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Disassembly
Register Va
Current
R0
R1
R2
R3
R4
R5
R6
R7
R8
R9
R10
R11
R12
R13 (SP)
R14 (LR)
R15 (PC)
CPSR
SPSR
User/System
Fast Interrupt
Interrupt
Supervisor
Abort
24: PINSEL0 &= ~(3 << 20);
25:
26: // Configure P0.10 as output
0x0000019C E59F0050 LDR R0,[PC,#0x0050]
nine.c
18 T0IR = 0x01; // Clear interrupt flag
19 }
20
21 int main()
22 {
23 // Configure P0.10 as GPIO
24 PINSEL0 &= ~(3 << 20);
25
26 // Configure P0.10 as output
27 IO0DIR |= (1 << 10);
28
29 while(1)
30 {
31 IO0SET = (1 << 10); // LED ON
32 timer0_delay(1000); // 1 second delay
33
34 IO0CLR = (1 << 10); // LED OFF
35 timer0_delay(1000); // 1 second delay
36 }


U1
X1
CRYSTAL
C1
1nF
C2
1nF
+3.3
R2
10k
R1
330
D1
LED-RED
XTAL1
XTAL2
RTXC1
RTXC2
RST
VBAT
VREF
V3A
V3
V3
VSSA
VSS
VSS
VSS
VSS
VSS
P0.0/TxD0/PWM1
P0.1/RxD0/PWM3/EINT0
P0.2/SCL0/CAP0.0
P0.3/SDA0/MAT0..0/EINT1
P0.4/SCK0/CAP0.1/AD0.6
P0.5/MISO0/MAT0.1/AD0.7
P0.6/MOSI0/CAP0.2/AD1.0
P0.7/SSEL0/PWM2/EINT2
P0.8/TxD1/PWM4/AD1.1
P0.9/RxD1/PWM6/EINT3
P0.10/RTS1/CAP1.0/AD1.2
P0.11/CTS1/CAP1.1/SCL1
P0.12/DSR1/MAT1.0/AD1.3
P0.13/DTR1/MAT1.1/AD1.4
P0.14/DCD1/EINT1/SDA1
P0.15/RI1/EINT2/AD1.5
P0.16/EINT0/MAT0.2/CAP0.2
P0.17/CAP1.2/SCK1/MAT1.2
P0.18/CAP1.3/MISO1/MAT1.3
P0.19/MAT1.2/MOSI1/CAP1.2
P0.20/MAT1.3/SSEL1/EINT3
P0.21/PWM5/AD1.6/CAP1.3
P0.22/AD1.7/CAP0.0/MAT0.0
P0.23
P0.25/AD0.4/AOUT
P0.26/AD0.5
P0.27/AD0.0/CAP0.1/MAT0.1
P0.28/AD0.1/CAP0.2/MAT0.2
P0.29/AD0.2/CAP0.3/MAT0.3
P0.30/AD0.3/EINT3/CAP0.0
P0.31
P1.16/TRACEPKT0
P1.17/TRACEPKT1
P1.18/TRACEPKT2
P1.19/TRACEPKT3
P1.20/TRACESYNC
P1.21/PIPESTAT0
P1.22/PIPESTAT1
P1.23/PIPESTAT2
P1.24/TRACECLK
P1.25/EXTIN0
P1.26/RTCK
P1.27/TDO
P1.28/TDI
P1.29/TCK
P1.30/TMS
P1.31/TRST
LPC2138